

Projet virtualisation - Shortener

E5FI - Gian Franco SPADARO et Raphaël ORSI

Liens vers le projet : <https://github.com/raph-o/e5fi-virtualisation>

Labs

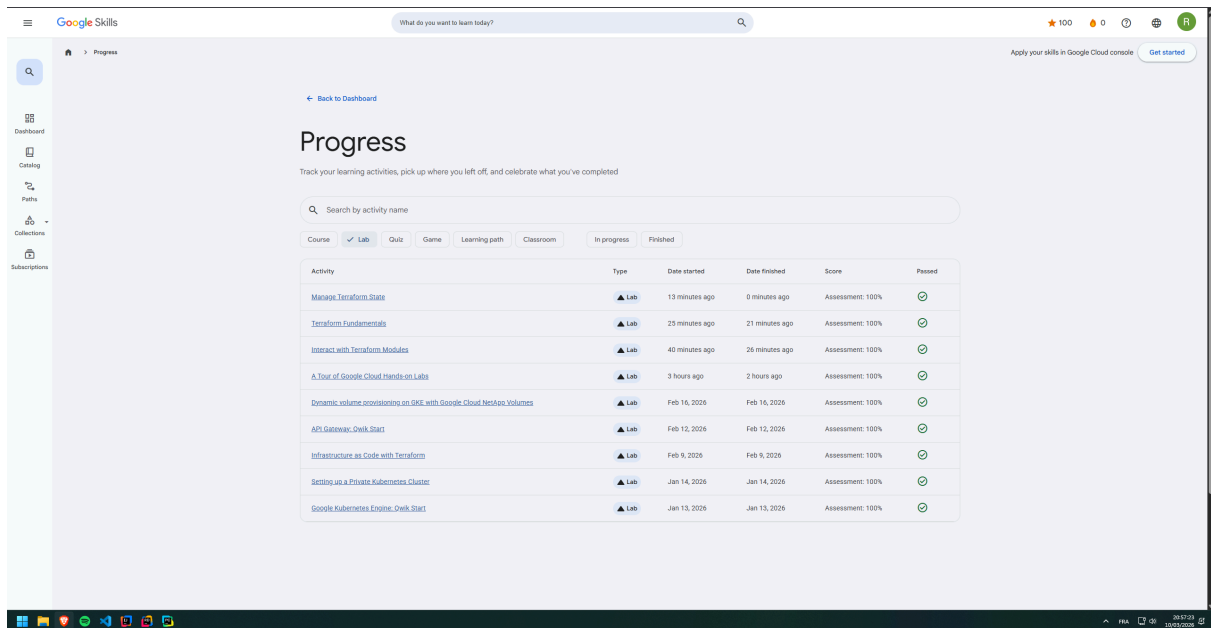
Gian

Activity	Type	Date started	Date finished	Score	Passed
Interact with Terraform Modules	▲ Lab	19 minutes ago	0 minutes ago	Assessment: 100%	✓
Terraform Fundamentals	▲ Lab	30 minutes ago	19 minutes ago	Assessment: 100%	✓
Dynamic volume provisioning on GKE with Google Cloud NetApp Volumes	▲ Lab	57 minutes ago	36 minutes ago	Assessment: 100%	✓
API Gateway: Qwik Start	▲ Lab	1 hour ago	1 hour ago	Assessment: 100%	✓
Infrastructure as Code with Terraform	▲ Lab	2 hours ago	1 hour ago	Assessment: 100%	✓
Build Infrastructure with Terraform on Google Cloud	📖 Course	2 hours ago			
Getting Started with Google Cloud	📖 Path	3 hours ago			
A Tour of Google Cloud Hands-on Labs	▲ Lab	3 hours ago	2 hours ago	Assessment: 100%	✓
Setting up a Private Kubernetes Cluster	▲ Lab	Jan 14, 2026	Jan 14, 2026	Assessment: 100%	✓
Google Kubernetes Engine: Qwik Start	▲ Lab	Jan 14, 2026	Jan 14, 2026	Assessment: 100%	✓

Labs bonus:

- [Interact with terraform modules](#),
- [Terraform fundamentals](#)

Raphaël



Labs bonus:

- [Interact with terraform modules](#),
- [Terraform fundamentals](#)
- [Manage terraform state](#)

Structure

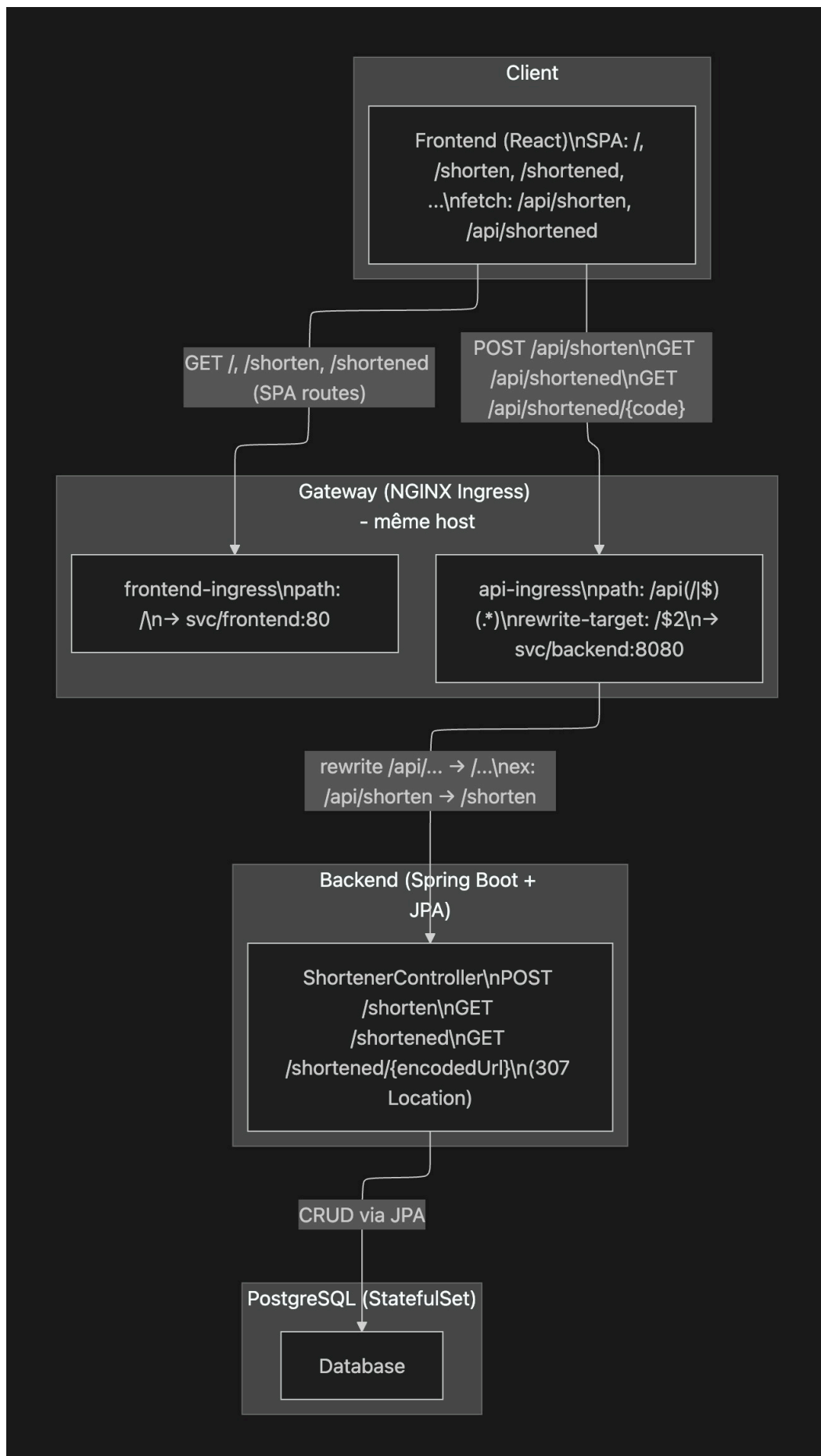
- backend: api
- frontend: interface
- k8s: fichiers de déploiement du projet sous kubernetes
- build-images.sh: script pour rapidement build les images docker
- cleanup.sh: script pour rapidement supprimer l'infrastructure du projet dans minikube
- deploy.sh: script pour rapidement déployer tous le projet dans minikube
- contract.yml: contrat de l'api

Technologies

- backend: Java 21, Spring Boot 3.2, Spring Data JPA, Gradle 8.4
- frontend: React 19, Vite 7.2
- bdd: H2 en local / PostgreSQL quand déployé

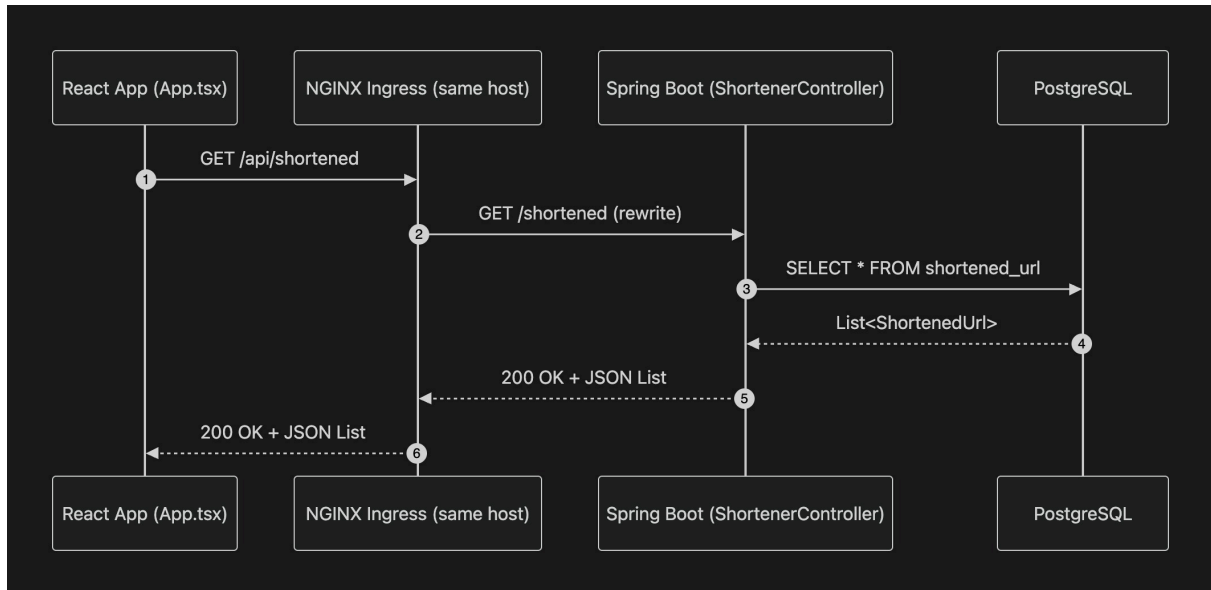
- gateway: NGINX Ingress Controller
- orchestration: Kubernetes (Minikube)
- containerisation: Docker

Architecture

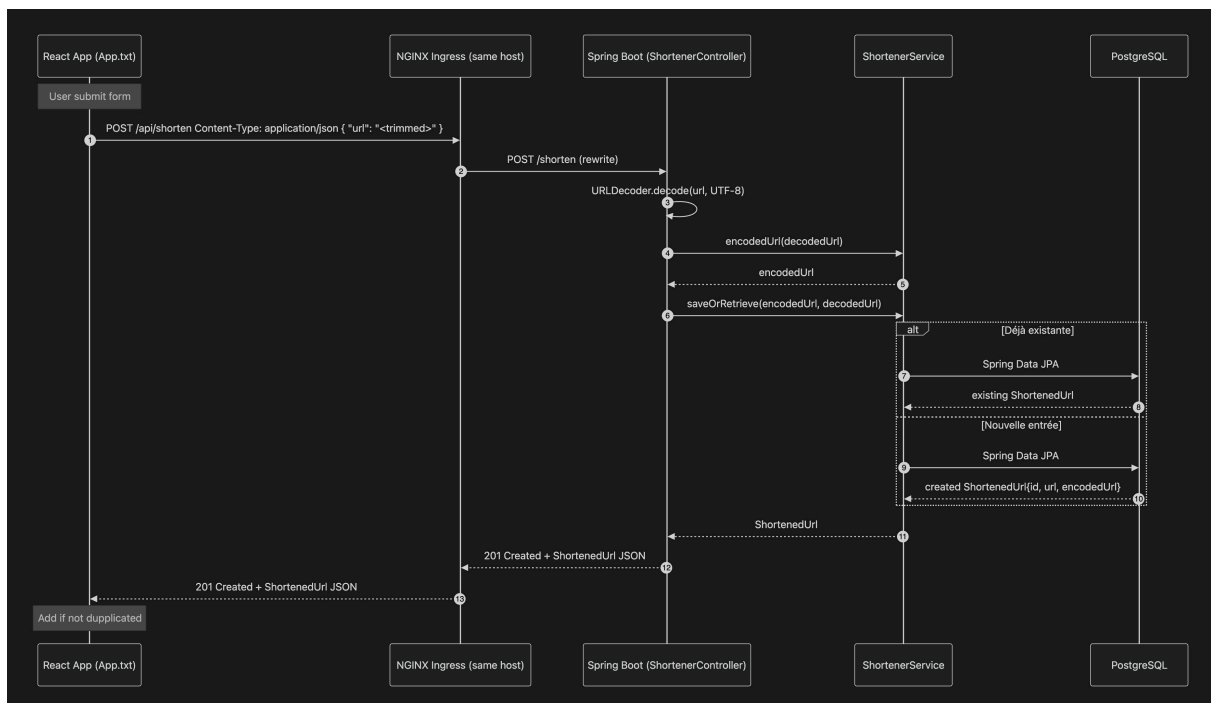


Flux

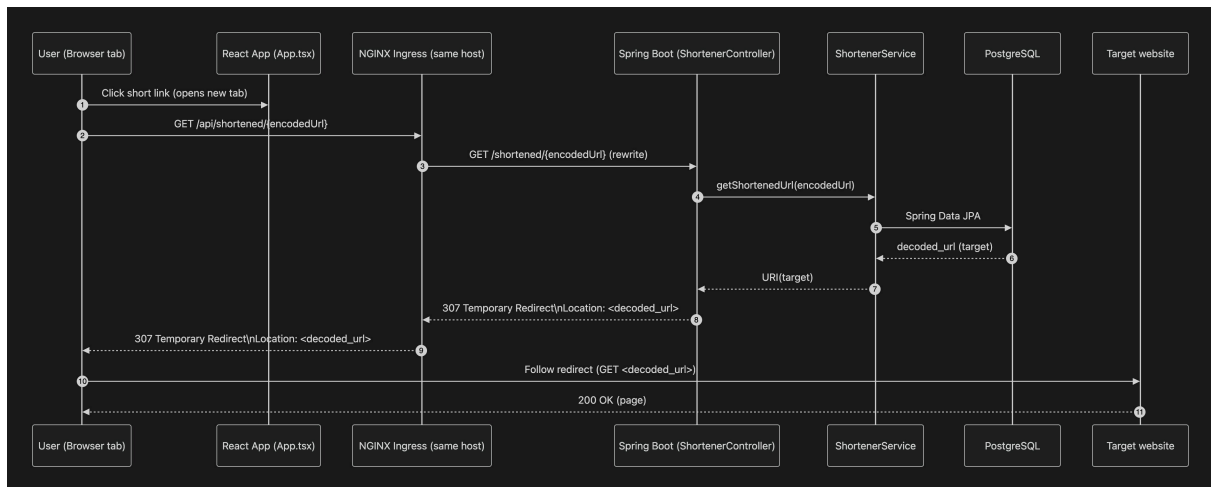
Liste des url raccourcies



Raccourcir une url



Accéder à l'url d'origine



Démarrage de l'application

Pré-requis

- Minikube installé
- Docker installé et actif

Déploiement

```
chmod +x deploy.sh
```

```
# Build les images et les déploie sur minikube
./deploy.sh
```

```
# Dans un nouveau terminal, ne pas quitter le processus
kubectl port-forward -n ingress-nginx svc/ingress-nginx-controller 8000:80
```

```
# Accéder à l'application
open http://localhost:8000
```

Nettoyage

```
chmod +x cleanup.sh
```

```
# Suppression de l'infrastructure dans minikube
./cleanup.sh
```

Interface

Mes URLs Raccourcies

Raccourcir

ID	Original	Raccourci (Cliquable)
1	https://google.com	lxJcG0jH

Mes URLs Raccourcies

Raccourcir

ID	Original	Raccourci (Cliquable)
1	https://google.com	lxJcG0jH
2	https://youtube.com	19riXlqwX